

Fantastical AI vs Grounded AI

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2012 ALEXNET



PERCEPTION AI

SPEECH RECOGNITION
DEEP RECSYS
MEDICAL IMAGING

GENERATIVE AI

DIGITAL MARKETING
CONTENT CREATION

AGENTIC AI

CODING ASSISTANT
CUSTOMER SERVICE
PATIENT CARE

PHYSICAL AI

SELF-DRIVING CARS
GENERAL ROBOTICS



Fantastical AI vs Grounded AI

Grounded A.I. in Urban Transportation

**Transport for London
(TfL)**

2005~2021

**Hong Kong
(MTR)**

2012~2019

**NRF, LTA, Singapore
(SMART)**

2016~2021, 2023~Present

**Chicago Transit Authority
(CTA)**

2001~2010, 2016~Present

**Boston
(MBTA)**

2003~2005, 2010~Present

**Washington DC
(WMATA)**

2021~Present

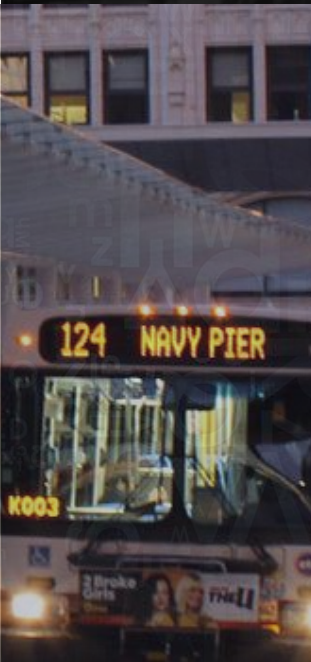


Control



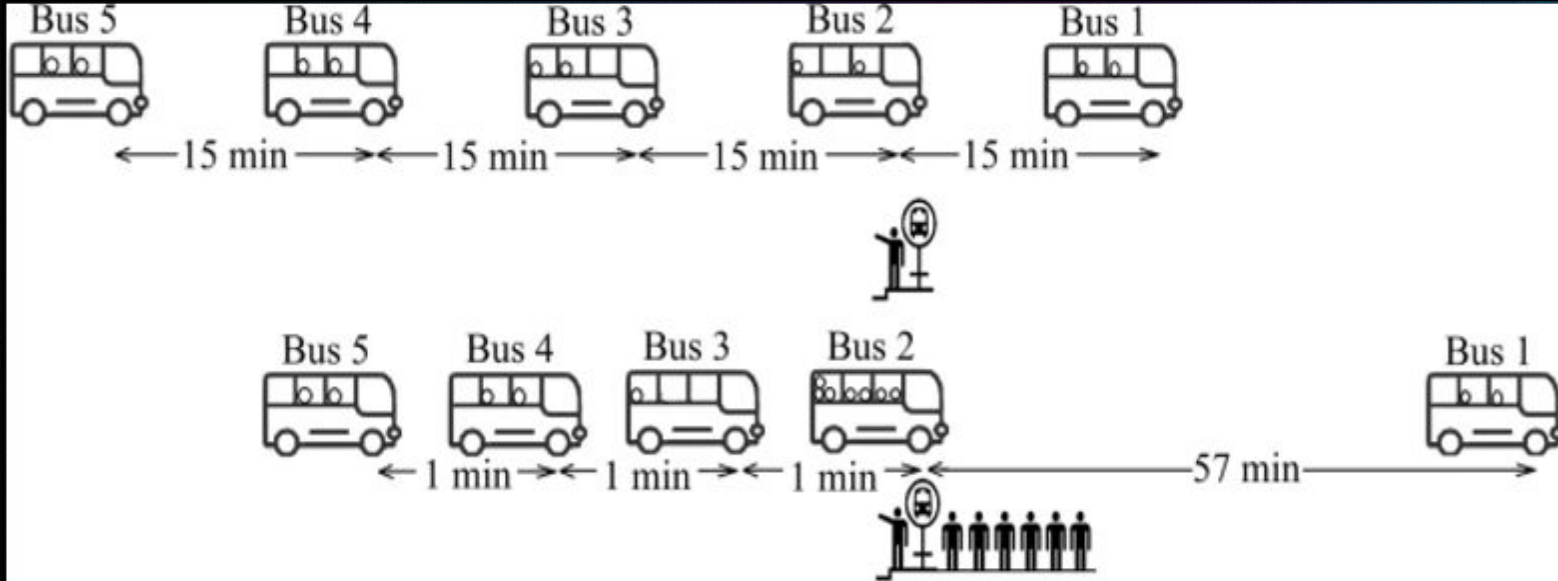
Assist

AI for Bus Dispatching to Improve Reliability





Bus bunching



Transit agencies experience major service reliability issues caused by driver shortage and driver absenteeism

TransitCenter |

Bus Operators in Crisis

**The Steady Deterioration of One of Transit's Most Essential Jobs,
and How Agencies Can Turn Things Around**

**CTA Service
delivered: 84%**

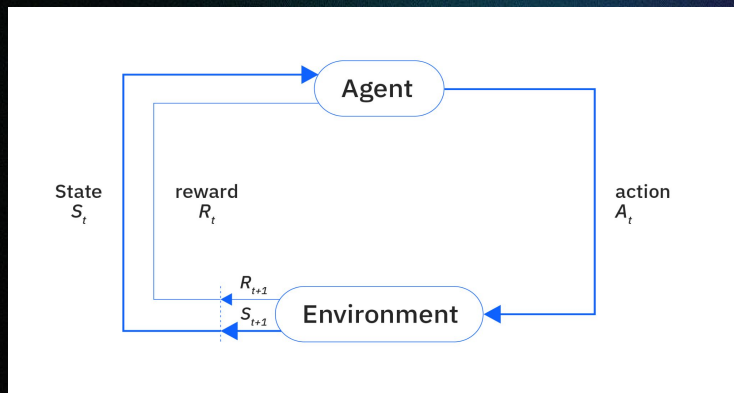
Improve service reliability through bus control using reinforcement learning

Robustness to uncertainties:

- Demand uncertainty: # of passengers
 - Supply uncertainty: driver absenteeism
-
- Environment uncertainty: traffic conditions
 - Operation uncertainty: non-compliance
 - Information uncertainty: Incomplete or inaccurate

Reinforcement learning

Actions not only affect the immediate reward, but also the next situation and all subsequent rewards.

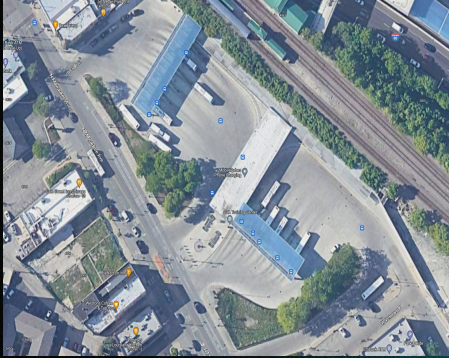


Control strategy at the bus terminal,

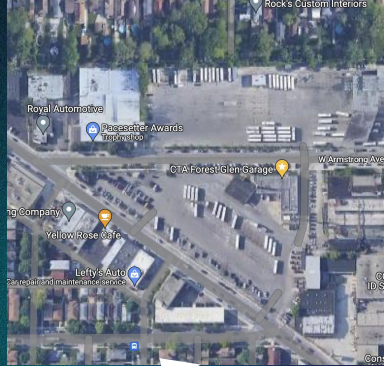
- Hold the bus by 1 min, 2 mins, 3mins, ...
- Move forward by 1 min, 2 mins, 3 mins,...

Human in the loop #1: Bus route 81 (actual) vs. 20 (initial)

Terminal: Jefferson Park,
serves 10 routes

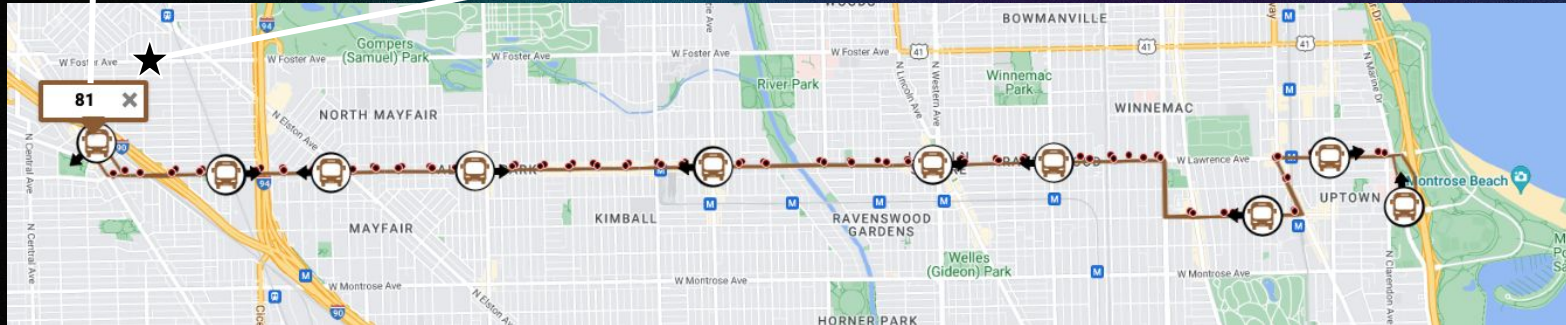


Garage: Forest Glen,
North Chicago routes

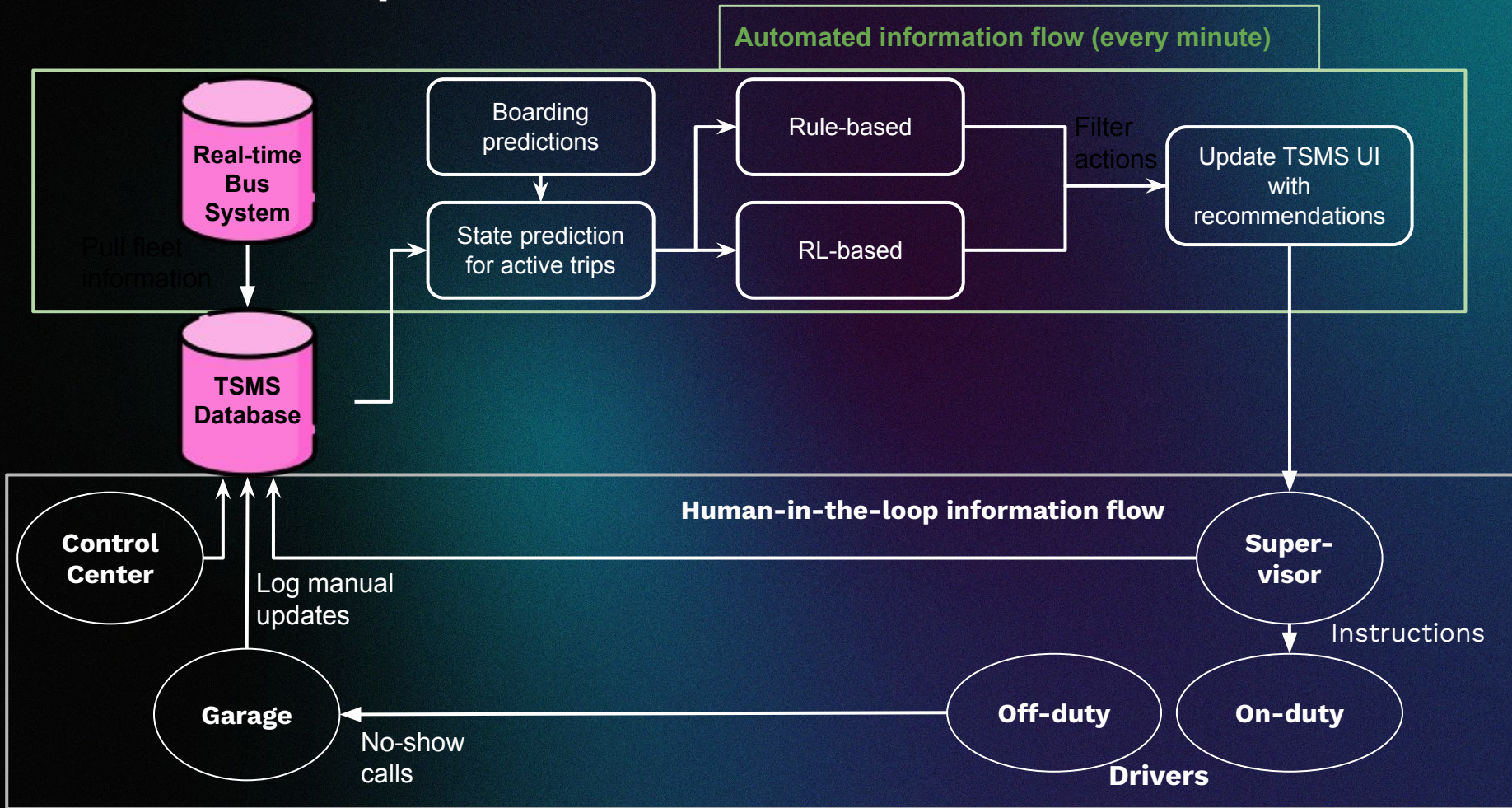


Route:

- 4 transfers to subway
- 50 stops each direction
- AM peak: 6-9 mins
- 15-25% missing trips
- 7,500 riders per weekday



Human in the loop #2: AI recommendation → Human control execution



Human in the loop #3: Field experiment in Chicago

Substantive support from the Chicago Transit Authority (CTA)

- President and CAA
- VP operation
- Garage supervisors
- Drivers



MIT developed the real-time control TSMS database, algorithm, and user interface

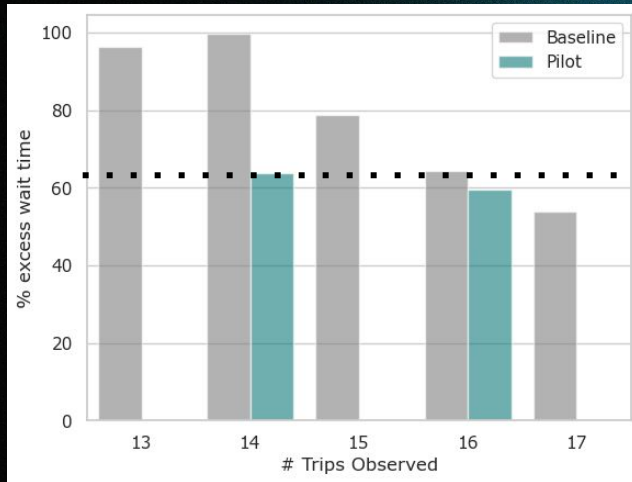
Impacts

Excess Waiting Time: reduced >30%

Impacts of Our Control Strategy: Performance Evaluation

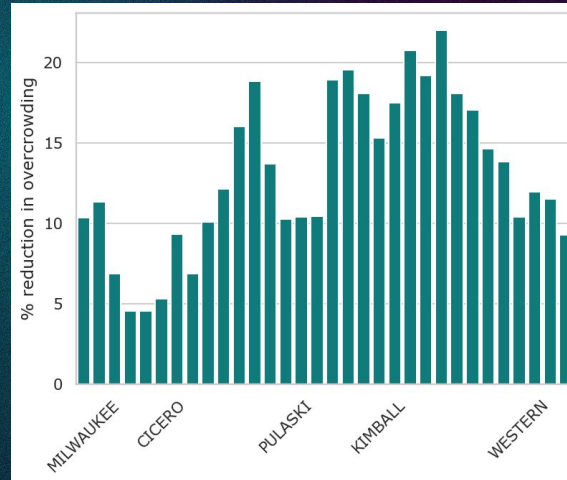
Access wait time by # of bus trips in AM period (6-9am)

37% reduction,
equivalent to adding 2 trips



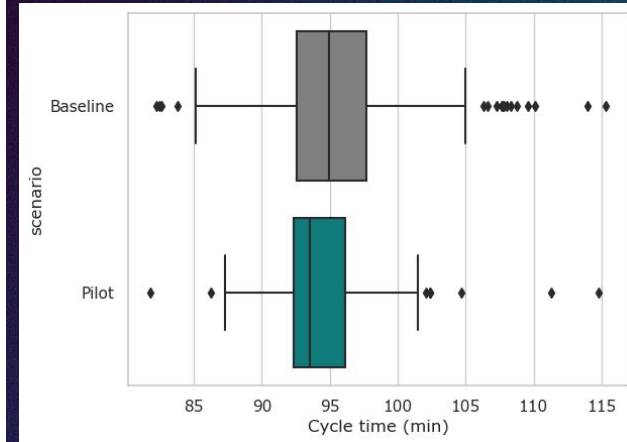
Overcrowding

5-20% reduction



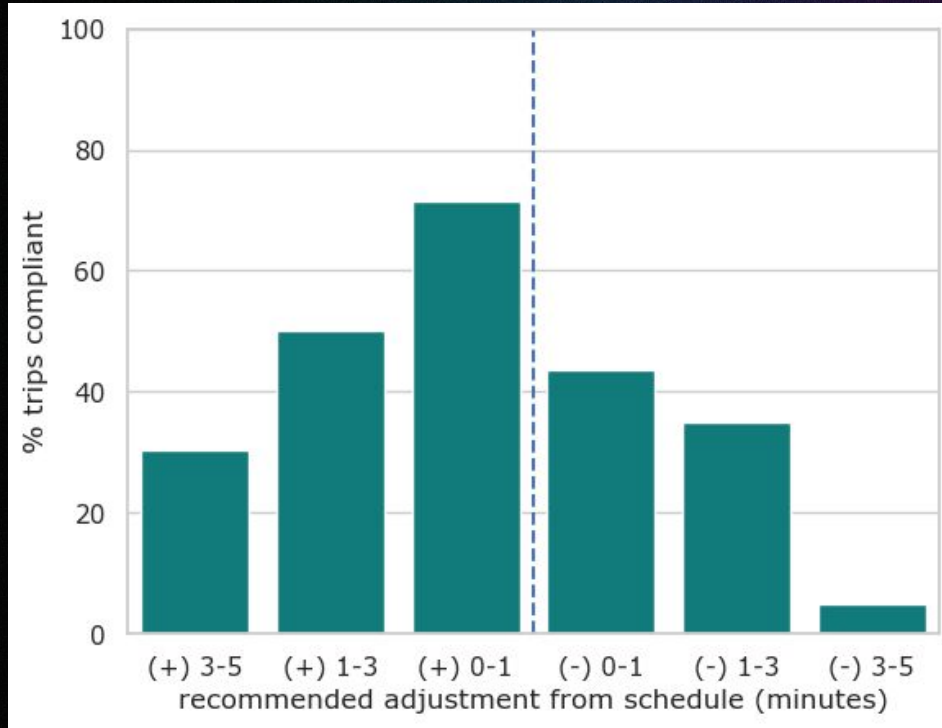
90th Cycle Times

2.5 minutes reduction
(2.4%) reduction



Do drivers follow the AI guidance?

Human in the loop #4: Do drivers follow A.I.?

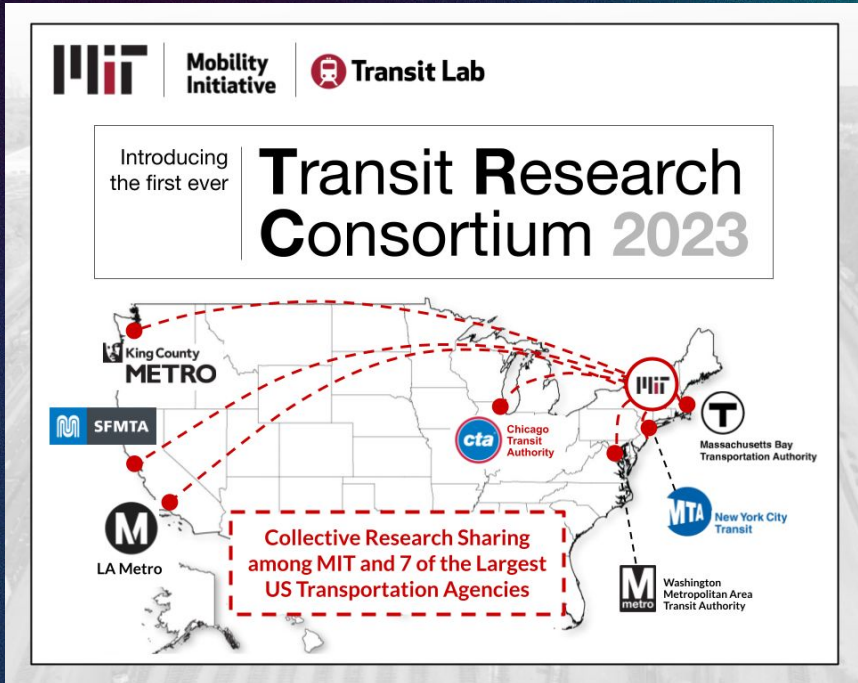


Driver compliance decreases with the aggressiveness of the recommended control actions.

Knowledge Transfer: MIT → Chicago

Category	Pilot (BP2)	Full-scale (BP3)
Time period	1 week	1 month
Routes	81	All routes at the Jefferson Park terminal
Control actions	Terminal dispatch	Mid-route holding, dynamic interlining, short-turning
Implementation	RAs overseeing operations	CTA staff takes over the software

Scaling Up: the US transit industry at large



Human in the loop

- Supervisor level
- Garage level
- Organizational level

Fantastical AI vs Grounded AI

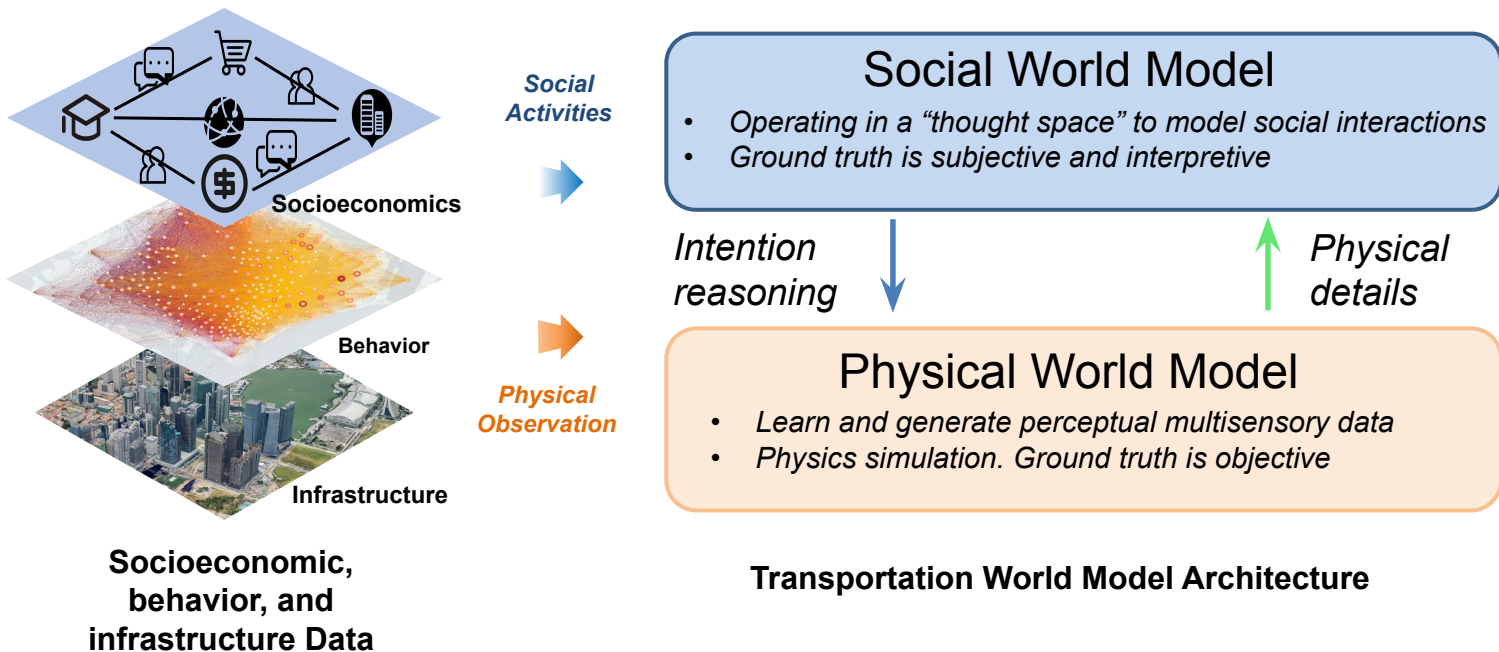
Fantastical AI: Ex 1

Transportation World Model

Transportation World Model: Bridging Physical and Social Dynamics

Multimodal Transport World Model:

understand and simulate both physical activity and social interaction.

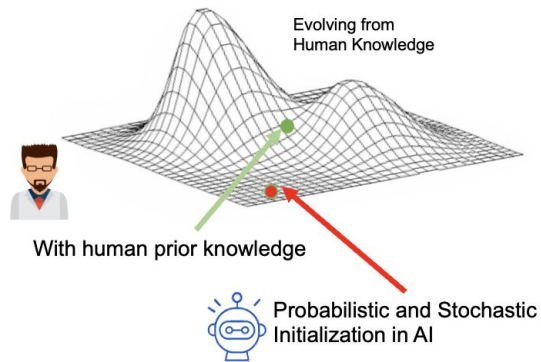


Fantastical AI: Ex. 2

AI for transportation science

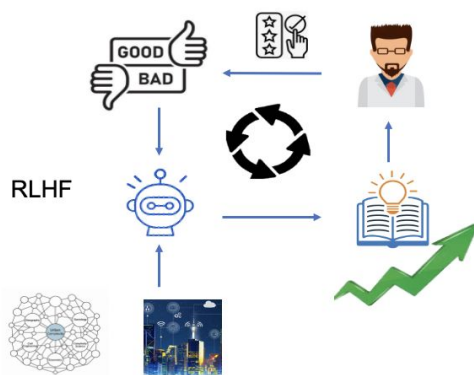
Agentic AI for Automated Transportation Science Discovery

Knowledge Integration



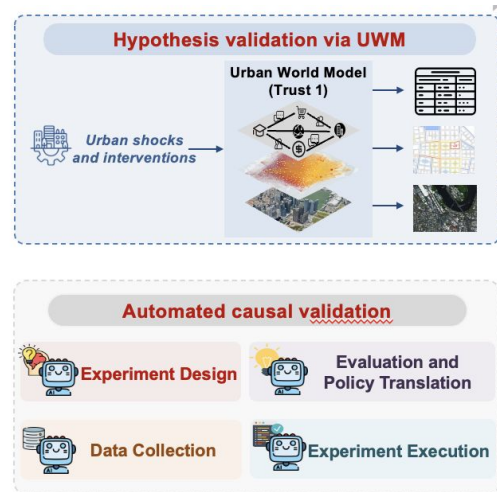
Integrate Human knowledge and physical representation from transport world model

Hypothesis Generation



Broader Scientific Hypothesis Space
e.g., Symbolic regression + LLM RLHF for discovery

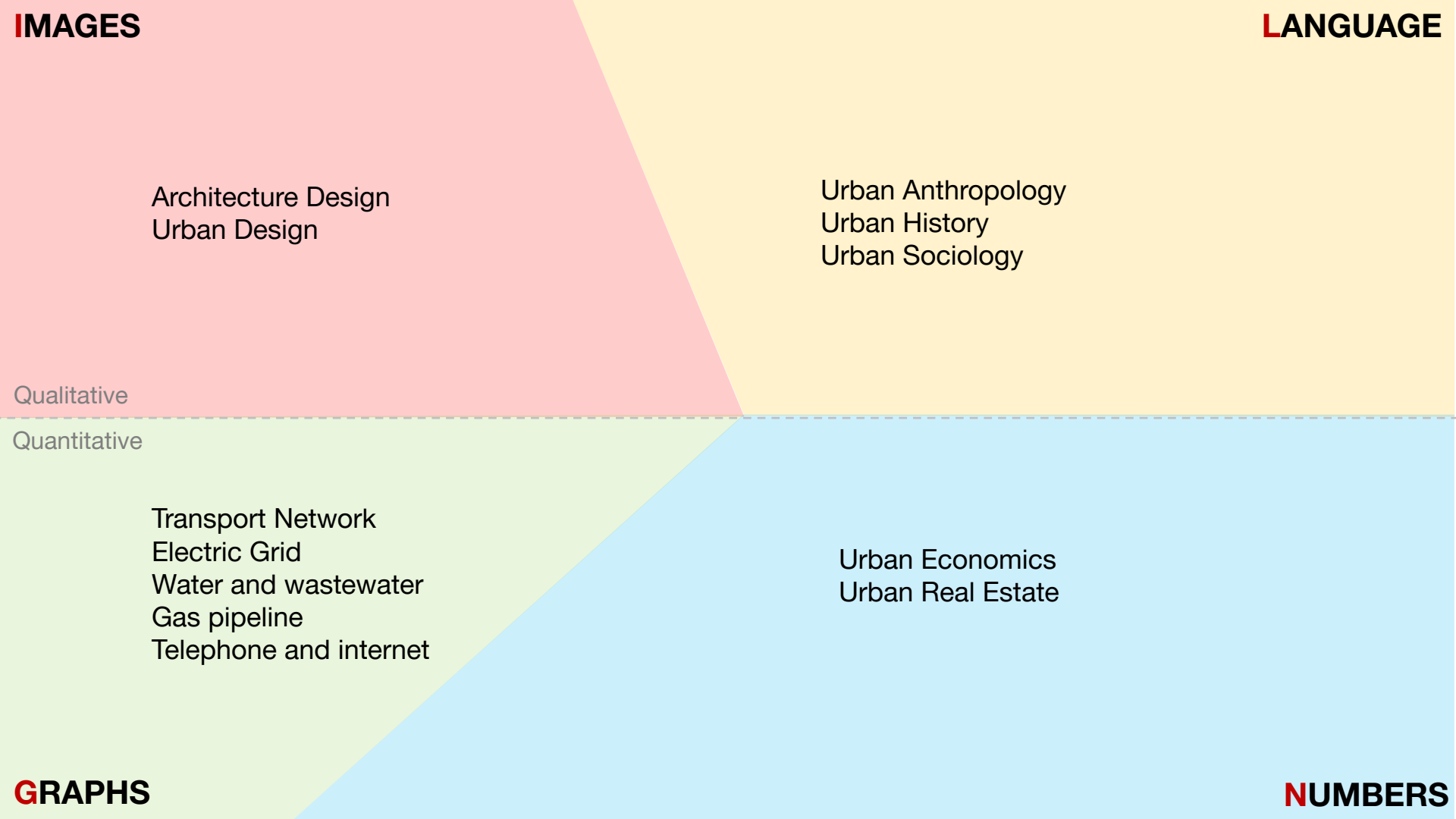
Scientific Validation



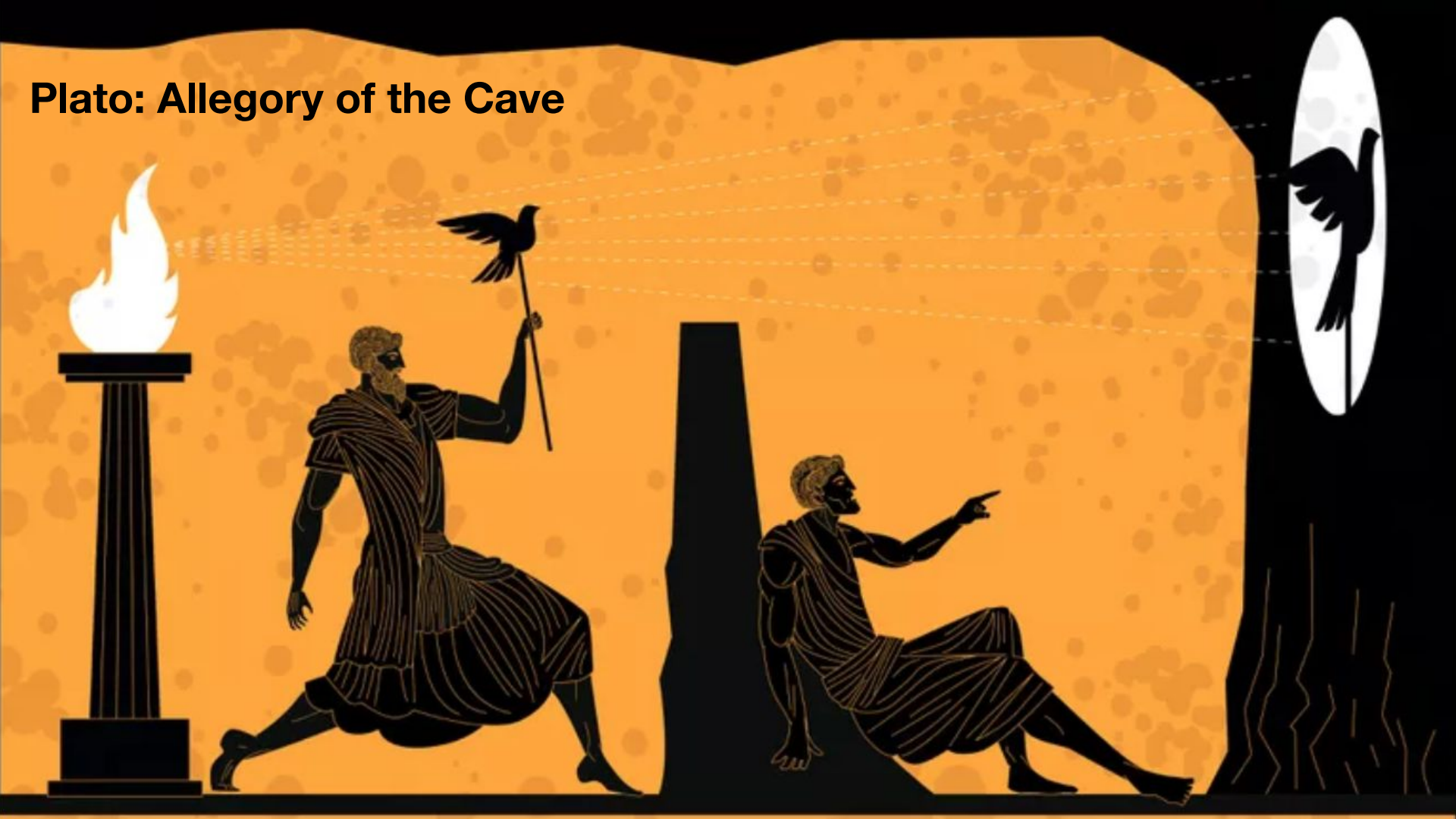
Automated Causal Validation
(world model as a sandbox)

Fantastical AI: Ex. 3

Multi-channel view of cities



Plato: Allegory of the Cave

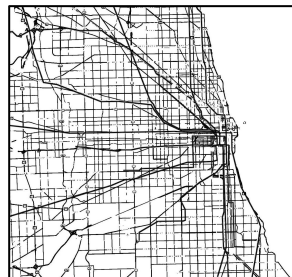


A multi-channel view city (e.g. Chicago)

Numeric values

Population : 2,693,976
Auto ownership per HH:
1.12/HH
...

Graphs (Transit and Road)



Images (Aerial or Street-view)



Natural Languages (From Wikipedia)

Chicago, officially the City of Chicago, is the most populous city in the U.S. state of Illinois, and the third-most-populous city in the United States. With an estimated population of 2,693,976 in 2019, it is also the most populous city in the Midwestern United States. Chicago is the county seat of Cook County, the second-most-populous county in the US, with a small portion of the northwest side of the city extending into DuPage County near O'Hare ...

Allegory of the Cities

cities projected to **graph**

cities projected to **image**cities projected to natural **language**

cities projected to **numbers**

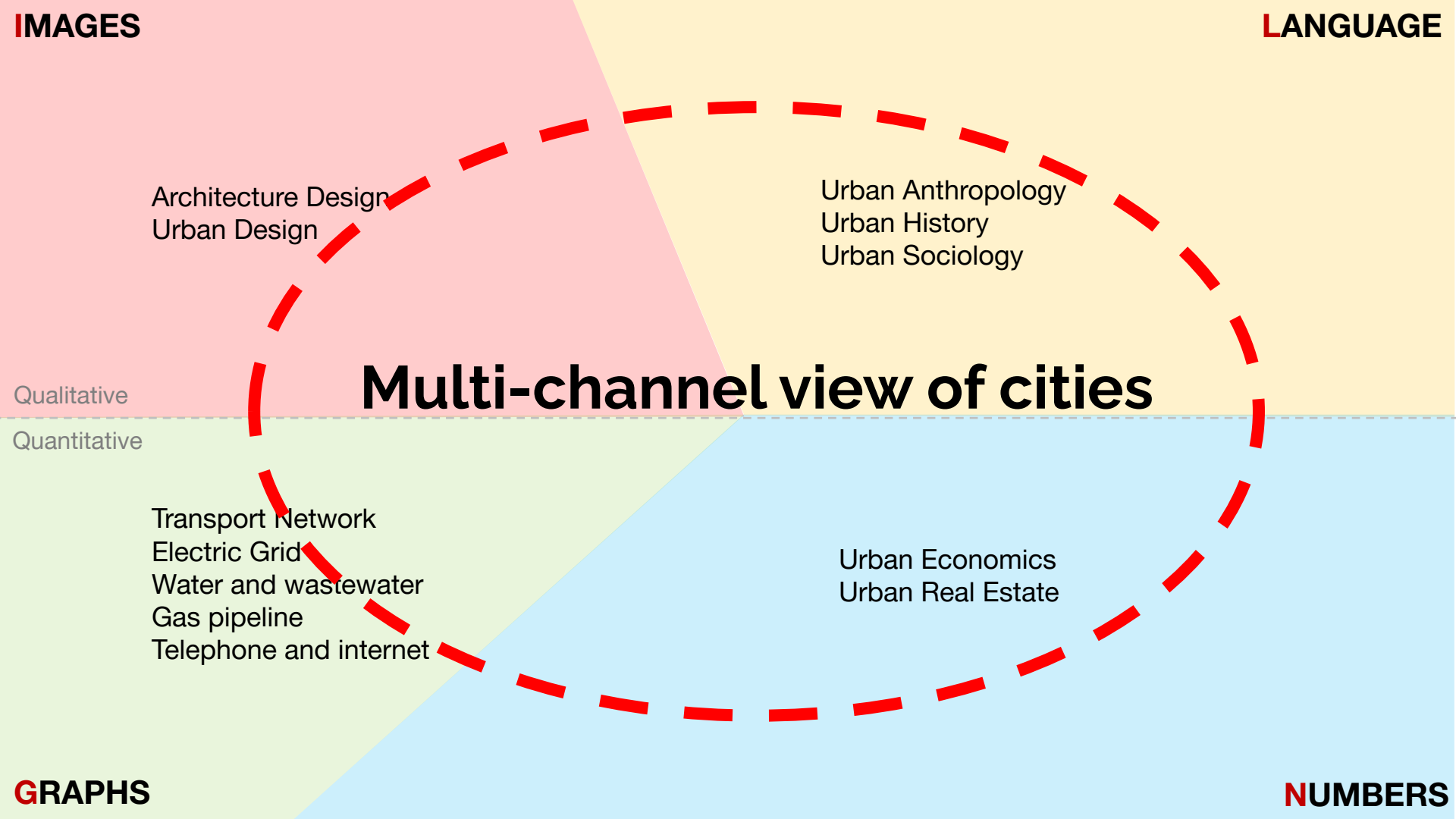


DNN: a unified framework for Graphs, Language, Images and Numbers

$$P(y_i|x_i) = f(x_i)$$

$$P(y_i|x_i, \text{Image}, \text{Text}, \text{Graph}) = f(x_i, \text{Image}, \text{Text}, \text{Graph}) \quad [\text{GLIN}]$$

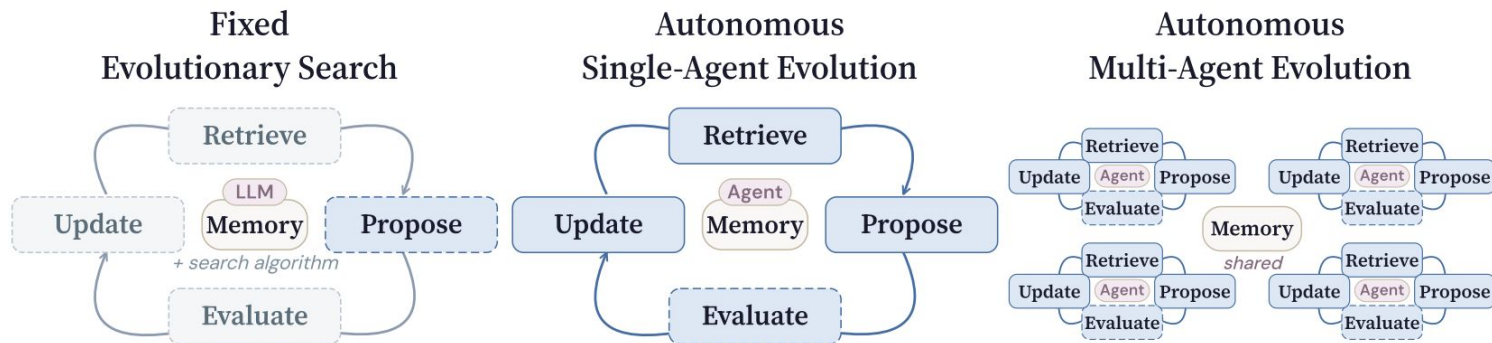
Diagram illustrating the unified framework for Graphs, Language, Images, and Numbers (GLIN). The equation shows the probability $P(y_i|x_i)$ is equal to the function $f(x_i)$. The input x_i is a vector that combines information from four modalities: Numbers, Language, Images, and Graph. The modalities are represented by icons: a blue cube for Numbers, a cityscape for Images, the text "We love Chicago" for Language, and a graph for Graph. Dashed arrows point from the modality labels to their respective icons. The output y_i is represented by a red cube icon.



Fantastical AI: Ex. 4

Recursive Self Improvement (RSI)

CORAL: Autonomous Multi-Agent Evolution for Open-Ended Discovery



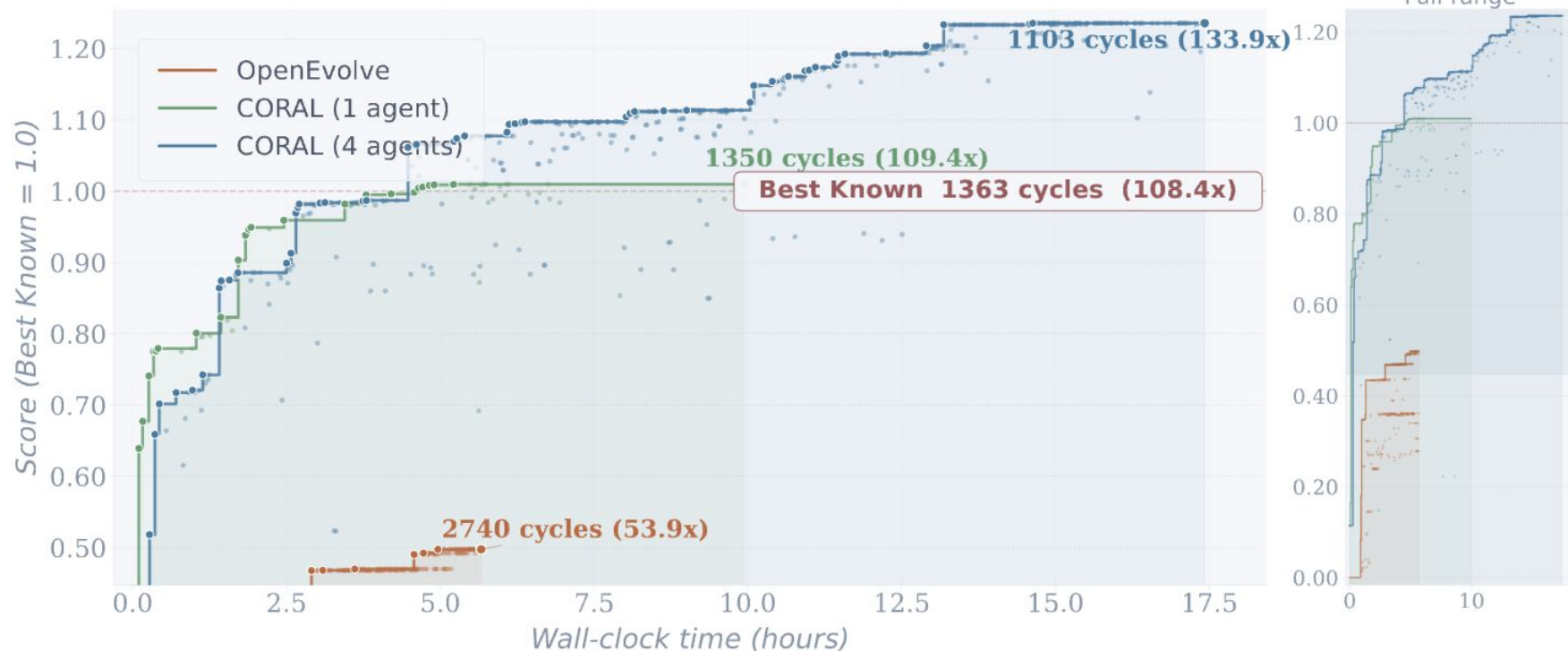
Retrieve	Context selected by fixed rules.	Agent chooses what to inspect.	Agent reads from shared memory.
Propose	Model proposes from given context.	Agent plans, implements, and tests.	Multiple agents propose in parallel.
Evaluate	Triggered by a fixed loop.	Agent decides when to evaluate.	Each agent schedules eval.
Update	Updated by fixed rules.	Agent decides what to store.	Agents write to shared memory.

Fixed
Partial autonomy
Full autonomy

Paradigms for LLM-based open-ended discovery

CORAL: Autonomous Multi-Agent Evolution for Open-Ended Discovery

CORAL outperforms previous SOTA on Anthropic's kernel engineering task by 20%



Fantastical AI vs Grounded AI in Transportation

Grounded

- Localized implementation
- Human behavioral reality
- Organizational constraints
- In-house capability

Fantastic

- Transportation World Model
- AI for Transportation Science
- Multi channel view of Cities
- Recursive Self Improvement (RSI)

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